

# A Novel Application of Ant Colony Optimization to $k$ -defective Maximum Edge Bicliques

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The maximum  $k$ -defective edge biclique ( $k$ -MDB) problem is an emerging NP-hard optimization. We adapt Ant Colony Optimization (ACO) to this problem and introduce a smaller side preference to outperform previously proposed heuristics. In some instances, ACO produces bicliques with 3-4x more edges than previously proposed heuristics. Additionally, ACO provides a tighter upper bound on the size of the  $k$ -MDB and can reduce brute force search time by over 90%, though compute constraints limit the robustness of our test. This work examines a new heuristic approach to the  $k$ -MDB problem while identifying several improvements for future heuristics.

## 1 Introduction

A bipartite graph is a common graph structure where one set of vertices is connected solely to another. Bipartite graphs can model user-product behavior, person-place interaction, and more. A biclique is a set of vertices on a bipartite graph such that each vertex on one side is connected to each vertex on the other. The maximum edge biclique problem aims to identify the biclique with the most edges on a bipartite graph. The problem has applications in online recommendation systems, fraud detection, and community detection [1].

Real-world data often introduces noise to graphs, and structures with relatively few missing connections may still be relevant. A biclique that is allowed up to  $k$  missing edges, or  $k$ -defective biclique ( $k$ -DB), can overcome this real-world noise. Identifying the maximum  $k$ -defective edge biclique ( $k$ -MDB) remains an emerging field.

The  $k$ -MDB problem is NP-hard. Unless  $P = NP$ , there is no algorithm that can run in polynomial time and is guaranteed to find the maximum solution. When the absolute optimum is not necessary, approximation techniques are useful to significantly reduce computation while providing meaningfully large bicliques.

Dorigo et al. [2] proposed a metaheuristic for exploring graphs known as Ant Colony Optimization (ACO). Initially used on identifying the shortest path in problems such as the Traveling Salesman Problem, ACO has been adapted

to many graph problems, such as vehicle routing [5], quadratic assignment [6], graph coloring [7], and the maximum clique problem [8], the last of which is closely related to the  $k$ -MDB problem via the reduction discussed in Section 2.2. Several improvements have been proposed to the algorithm since its inception, most notably the MAX-MIN Ant System (MMAS) [4]. We adapt MMAS as a heuristic approach to identifying the  $k$ -MDB. We observe that it performs significantly faster than brute-force algorithms while still returning comparably large  $k$ -defective bicliques.

## 2 Background

### 2.1 Definitions

Define a bipartite graph as a graph  $G = (U, V, E)$ , where  $U$  and  $V$  are two disjoint sets of vertices, and  $E \subseteq U \times V$  is the set of edges between  $U$  and  $V$ . Define  $n = |U| + |V|$  and  $m = |E|$ . Let  $S = (U_S, V_S)$  denote a subset of vertices where  $U_S \subseteq U$  and  $V_S \subseteq V$ . Let  $G(S) = (U_S, V_S, E_S)$  be the induced subgraph of  $S$  where  $E_S = \{(u, v) \mid u \in U_S \wedge v \in V_S \wedge (u, v) \in E\}$ . Define  $\bar{E}(S) = \{(u, v) \mid u \in U_S \wedge v \in V_S \wedge (u, v) \notin E\}$  be the set of non-edges in a subgraph. Let a  $k$ -defective biclique be a subgraph  $S$  where  $|\bar{E}(S)| \leq k$ .

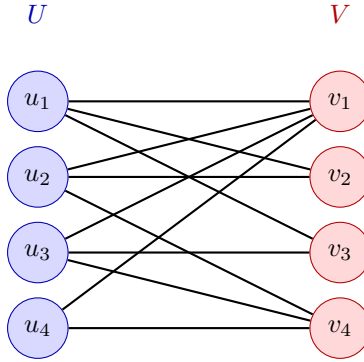


Figure 1: Example bipartite graph  $G = (U, V, E)$

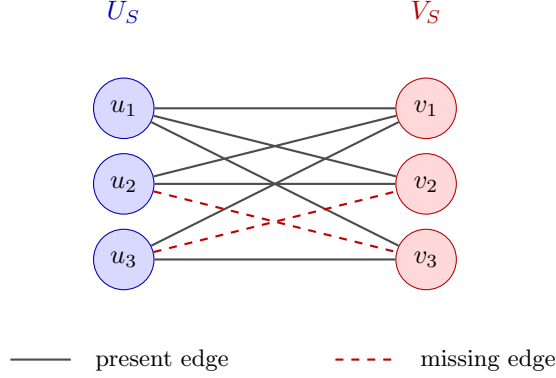


Figure 2: A 2-defective biclique present in Figure 1. Every vertex in  $U_S$  is connected to every vertex in  $V_S$  except for two missing edges. Since  $|\bar{E}(S)| = 2$ , this is a  $k$ -defective biclique, where  $k \geq 2$ .

Define the neighbors of a node  $u$  in a subgraph  $S$  as  $N_S(u) = \{v \in S \mid \{u, v\} \in E(S)\}$ . Define the nonneighbors of a node as  $\bar{N}_S(u) = \{v \in S \mid (u, v) \notin E(S)\}$ . Define the degree of a node as  $d_S(u) = |N_S(u)|$ , and the nondegree as  $\bar{d}_S(u) = |\bar{N}_S(u)|$ .

Define a maximal  $k$ -MDB as a subgraph  $S$  that cannot be expanded. That is, for any node  $u \notin S$ ,  $|\bar{E}(S \cup \{u\})| > k$ . Define the maximum  $k$ -MDB as the largest maximal  $k$ -MDB in a graph  $G$ . Let  $\theta$  be a user-defined limitation on the solution space. The algorithms proposed in this paper seek to find the maximum  $k$ -MDB where both sides have at least  $\theta$  vertices. This is because, on real-world graphs,  $k$ -defective bicliques can be formed with many edges but disproportionate vertex counts, where one side has very few vertices while the other has many [1]. Without the  $\theta$  term, the utility of the algorithm is limited.

## 2.2 Computational Complexity

Finding the maximum  $k$ -MDB on a bipartite graph is an NP-hard optimization problem. The problem reduces to the well-known NP-hard maximum clique problem, which seeks to find the maximum set of vertices on a graph connected to every other vertex [1].

## 2.3 Previous Work

Cui et al. [1] proposed a brute-force approach to the  $k$ -MDB problem, as well as an initial heuristic designed to find a  $k$ -MDB with exactly  $\theta$  nodes on one side of the graph.

Concurrently, Wang et al. [11] studied a related but distinct formulation: the maximum  $k$ -defective biclique measured by vertex count rather than edge count, with a similar minimum vertex count on both sides. They develop an exact branch-and-bound algorithm with time complexity  $\mathcal{O}^*(\gamma^{n+k})$ , tightened to

$\mathcal{O}^*(\gamma^{\alpha\delta^2+k})$  by exploiting a diameter-three property of large defective bicliques, and propose an analogous construct-and-expand heuristic (greedy initial construction followed by one-sided greedy expansion) to seed their search. Because their objective differs from ours, we do not benchmark directly against their algorithm, but their diameter-three pruning and expansion-based seeding strategy are directly relevant to future extensions of this work (Section 6.1).

## 3 Methods

### 3.1 ACO Algorithm

ACO traditionally has a set of ants that traverse a graph, leaving pheromone that attracts other ants to follow their path. As the ants explore the solution space, they all converge on one particular path as the quantity of pheromones on that path becomes increasingly larger.

We adapt the ACO algorithm to the  $k$ -MDB problem. A population of ants iteratively adds vertices from  $G$  to a subgraph  $S$ , preferring nodes with higher  $d_S$  and higher pheromone  $\tau$ . Here,  $\sigma(x) = \frac{1}{1+e^{-x}}$  denotes the standard logistic sigmoid, used to dampen the impact of a node’s degree with the entire graph as the size of the subgraph grows. Ants prefer to add nodes connected to the node they most recently visited. Intuitively, this reduces the number of nodes that an ant has to consider, and increases the likelihood that early on, an ant adds nodes that are properly connected to the subgraph.

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**Algorithm 1** ACO

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**Ensure:** a subset of vertices  $S$  such that  $G(S)$  is a  $k$ -MDB

```
function ACO( $G, N_e, N_S, \rho, T$ )  
  for all  $u \in G$  do  
     $\tau(u) \leftarrow 1$   
   $e \leftarrow 0$   
   $B \leftarrow \{\}$   
  while  $e < N_e$  do  
     $e \leftarrow e + 1$   
     $S \leftarrow N_S$  empty subgraphs  
     $\tau_C \leftarrow \tau$   
    parallel for each  $S_i \in S$  do  
       $C \leftarrow \{u \in G \mid k \geq |\bar{E}(S \cup \{u\})|\}$   
      while  $|C| > 0$  do  
         $L \leftarrow$  the last node that was appended to  $S$   
         $C_L \leftarrow d_L$   
        if  $|C \cap C_L| > 0$  then  
           $search \leftarrow C \cap C_L$   
        else  
           $search \leftarrow C$   
         $next \leftarrow$  linear sample of  $u \in search$  based on  
        DESIRABILITY( $u, S_i, \tau_C$ )  
        add  $next$  to  $S_i$   
         $\tau_C(u) \leftarrow \tau_C(u) + T$   
       $C \leftarrow \{u \in G \mid k \geq |\bar{E}(S \cup \{u\})|\}$   
     $S_B \leftarrow \operatorname{argmax}_{S_i \in S} \operatorname{INSTANCEFITNESS}(S_i) \triangleright S_B$  is guaranteed to have  
    at least one node  
    if  $B = \emptyset$  or  $\operatorname{INSTANCEFITNESS}(S_B) > \operatorname{INSTANCEFITNESS}(B)$  then  
       $B \leftarrow S_B$   
     $\tau \leftarrow \tau_C$   
    for all  $u \in G$  do  
       $\tau(u) \leftarrow \tau(u) \times (1 - \rho)$   
  return  $B$ 
```

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We observe that ACO has trouble generating graphs with at least  $\theta$  nodes because it often adds a disproportionate number of nodes on one side. We first introduce an instance fitness function that rewards a more balanced biclique by squaring the minimum number of nodes. For the desirability function, we introduce a hyperparameter  $P$ , a factor that favors nodes on the smaller side when one side of an ant's subgraph has  $\geq \theta$  nodes and another side has  $< \theta$ . We use the following desirability.

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**Algorithm 2** Prefer-Smaller-Side Desirability

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```
function INSTANCEFITNESS( $S$ )
     $a, b \leftarrow \text{minmax}(|U_S|, |V_S|)$ 
    if  $a \geq \theta$  then
        return  $b \times a^2$ 
    else
        return  $a^2$ 
function DESIRABILITY( $u, S, \tau$ )
     $\eta \leftarrow d_S(u) + d_G(u) \times \sigma(|U_S| + |V_S|)$ 
    if  $(|U_S| \geq \theta \oplus |V_S| \geq \theta)$  and  $u$  is on the smaller side then
         $\eta \leftarrow \eta \times P$  ▷ prefer node on the smaller side
    return  $\tau(u) \times \eta^3$ 
```

---

We attempted an elitist approach to lay additional pheromone on subgraphs with higher fitnesses according to a fitness function, as well as implementing local tabu search on ant subgraphs to prioritize solutions that are more likely to grow. We implemented the MAX-MIN Ant System as well. Finally, we attempted a genetic algorithm approach adapted from Shah’s GCLIQUE framework [3], but found it underperformed both ACO and the  $\theta$ -heuristic, likely due to the high dimensionality of the encoding required to represent biclique membership across both vertex sets.

### 3.2 Brute Force Approach

Finally, we implement the approach proposed by Cui et al. These authors observe that many  $k$ -MDB have one side with exactly  $\theta$  vertices and propose a heuristic that removes nodes from one side while adding nodes to another until there are  $\theta$  nodes on that side. We summarize their algorithm below.

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**Algorithm 3**  $\theta$ -Heuristic

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**Ensure:** a  $k$ -defective biclique of  $G$

```
function THETAHEURISTIC( $G$ )
     $U \leftarrow \{\}$ 
     $V \leftarrow V_G$ 
    while  $|U| < \theta$  do
        append  $u \in U_G \setminus U$  with largest  $d_V$  to  $U$ 
        while  $|\bar{E}(U \cup V)| > k$  do
            remove  $v \in V$  with largest  $\bar{d}_U$  from  $V$ 
    return  $U \cup V$ 
```

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They then propose a branch-and-pivot algorithm that takes advantage of this initial heuristic upper bound. The algorithm checks combinations of  $S = (U_S, V_S)$  by maintaining a candidate set  $C = (U_C, V_C)$  of all remaining nodes  $u \in G$  s.t.  $|\bar{E}(S \cup \{u\})| \leq k$  that could still be added to  $S$ . It recursively chooses

a node  $u \in C$  and branches on  $(S' = S \cup \{u\}, C' = \text{candidate set of } S')$  and  $(S' = S, C' = C \setminus \{u\})$ . Several optimizations are added to make this algorithm faster than a naive brute-force approach. Its time complexity is  $O(m\beta_k^n)$ , where  $\beta$  is the largest real root of the equation  $x^{2k+5} - 2x^{2k+4} + x^{k+3} - 2x + 2 = 0$ . See Appendix B for full pseudocode.

The paper proposes several novel optimization techniques as well. These techniques are beyond the scope of this paper but were implemented during testing. These optimizations include several graph reduction techniques that remove nodes provably not part of a  $k$ -MDB.

### 3.3 Time Complexity

TO-DO: Review this and make sure it makes sense, but other than that... idk

Let  $n = |U_G| + |V_G|$  and  $m = |E_G|$ . For each ant, it must first examine  $n$  nodes. At the next step, it will examine a maximum of  $n-1$  nodes, and then  $n-2$  nodes, etc. Therefore, an ant will examine at most  $n + (n-1) + (n-2) \dots = n(n+1)/2$ . Additionally, at each step, an ant generates a candidate set of nodes that could still be added to its subgraph. This pass is optimized to be  $\mathcal{O}(|U_C| + |V_C|)$ , where  $C$  is the set of candidate nodes before it advanced. If  $C$  remains very large, the upper time complexity would therefore be  $\mathcal{O}(A \times I \times (n^2 + n))$  for  $A$  ants advancing for  $I$  iterations, but this would be dominated by  $n^2$ . In a parallelized implementation with  $p$  processors, the true upper complexity bound is  $\mathcal{O}(\lceil A/p \rceil \times I \times n^2)$ .

In practice, however, there are two factors limiting this complexity. First, ants run until they are unable to add more nodes. Let  $D$  be the  $k$ -MDB of  $G$ . Because smaller side preference encourages subgraphs with  $\theta$  nodes on each side, an ant will empirically visit a maximum of  $n_D = |U_D| + |V_D|$  nodes, and therefore have a time complexity of  $\mathcal{O}(\lceil A/p \rceil \times I \times n_D \times n)$ .  $n_D$  is often much lower than the total number of nodes. Additionally, let  $L$  be the last node an ant visited. An ant prefers to visit only the intersection of  $d_L$  and its candidate set, so it often only visits  $d_L$  or fewer nodes. Therefore, the practical time complexity is  $\mathcal{O}(\lceil A/p \rceil \times I \times n_D \times (n + \max_{v \in G} d_v))$ . Because  $\max_{v \in G} d_v$  is necessarily lower than  $n$  and almost always lower than  $\min(|U_G|, |V_G|)$ , ACO's complexity practically does not scale linearly with  $n$  after the first iteration.

The  $\theta$ -heuristic has a time complexity of  $\mathcal{O}(\theta n + m)$  [1].

### 3.4 Experimental Setup

Unless otherwise noted, all ACO experiments use the following hyperparameter values, selected via preliminary tuning on a small subset of graphs:

Table 1: ACO hyperparameters used in all experiments

| Parameter                              | Value                     |
|----------------------------------------|---------------------------|
| Evaporation rate ( $\rho$ )            | 0.05                      |
| Pheromone deposit ( $T$ )              | 1                         |
| Number of epochs ( $N_e$ )             | 5                         |
| Number of ants ( $N_S$ )               | varied, 1–100 (see below) |
| Smaller-side preference factor ( $P$ ) | 2                         |
| $k$                                    | 2                         |
| $\theta$                               | 5                         |

When testing ACO and the  $\theta$ -heuristic, a biclique with  $k$  missing edges and  $\theta$  vertices on each side was injected into the graph.

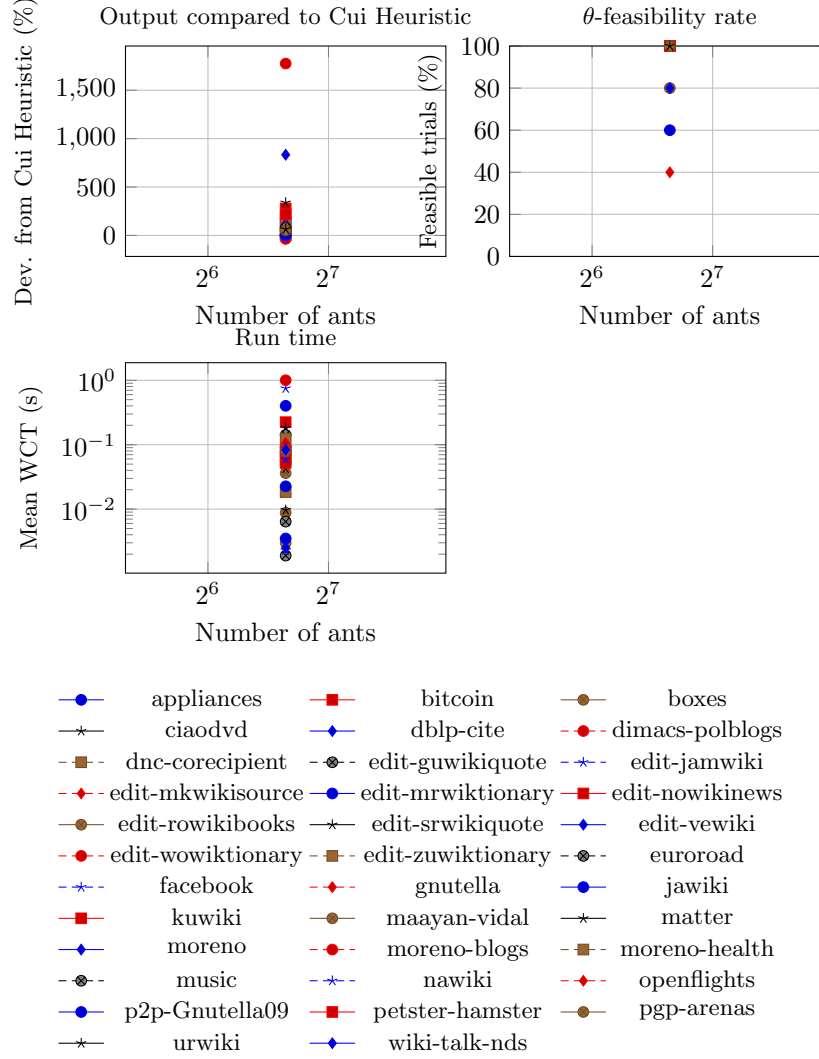
## 4 Results

We use several graphs from `konekt.cc`, a registry of bipartite graphs. We first applied the reductions proposed by Cui et al., and then ran both ACO and the  $\theta$ -heuristic.

All tests were run with 8 threads on a computer with Ubuntu, an Intel(R) Core(TM) Ultra 7 258V, and 32 gigabytes of RAM.

We also vary the number of ants and analyze the effect on run time as well as solution quality. We compare ACO results to  $\theta$ -heuristic results. Finally, because Branch and Pivot can prune more branches with a larger initial  $D^*$ , we demonstrate the speed increase from using ACO as the heuristic seeding method in cases where ACO outperformed the  $\theta$ -heuristic. Ant count  $N_S$  is varied across  $\{1, 2, 5, 10, 20, 50, 100\}$ .





We observe that a marginal increase in the initial upper bound of the sub-graph for Branch and Pivot allows the algorithm to prune more branches, and can sometimes result in significant decreases in runtime. ACO was run 5 times, and the largest  $k$ -DB was used. If this  $k$ -DB had more edges than the  $\theta$ -heuristic output, the pivot algorithm was then run using only the  $\theta$ -heuristic solution as a seed, and then using the ACO solution as a seed. We measure the difference between the pivot algorithm runtime, adding the time ACO took to run to its final pivot time. We summarize our results below.

Finally, we compared ACO and the  $\theta$ -heuristic on 32 graphs for a total of 5 iterations, and measured both the time to completion as well as the iterations and time to the best returned solution. We measured time and performance for

Table 2: Pivot seed comparison: ACO vs.  $\theta$ -heuristic

| Dataset           | Nodes | Edges | Reduced nodes | Reduced edges | Time saved (s) | Reduction (%) |
|-------------------|-------|-------|---------------|---------------|----------------|---------------|
| Maayan Vidal      | 4562  | 6749  | 215           | –             | 1.10           | 85.08         |
| Urwiki            | 2160  | 4018  | 59            | –             | 1.15           | 96.63         |
| Moreno Health     | 4754  | 12992 | 600           | –             | 1.29           | 79.30         |
| Edit Mkwikisource | 6853  | 13824 | 140           | 399           | 1.33           | 68.45         |
| Edit Zuwiktionary | 1618  | 7555  | 247           | –             | 52.76          | 50.28         |
| Edit Guwikiquote  | 2641  | 11387 | 778           | –             | 251.16         | 54.79         |
| Euroroad          | 1983  | 1440  | 10            | –             | -0.01          | -0.00         |
| Facebook          | 2913  | 3003  | 17            | –             | -0.02          | -0.00         |
| Moreno            | 2485  | 2300  | 25            | –             | -0.02          | -0.00         |
| Nawiki            | 1367  | 12108 | 408           | –             | -0.05          | -0.00         |
| Dnc Corecipient   | 1367  | 12108 | 408           | –             | -0.07          | -0.00         |
| Edit Srwikiquote  | 2225  | 8695  | 235           | –             | -0.21          | -0.01         |
| Kuwiki            | 1313  | 4951  | 0             | –             | -0.26          | -0.01         |
| Edit Vewiki       | 2075  | 11069 | 82            | –             | -0.26          | -0.01         |
| Edit Mrwiktionary | 4276  | 14375 | 75            | –             | -2.01          | -0.10         |
| P2P Gnutella09    | –     | 26036 | 561           | –             | -3.41          | -0.17         |

the  $\theta$ -Heuristic. In 17 trials, ACO produced a larger  $k$ -DB than the  $\theta$ -heuristic, underperforming in only 4 trials. In these trials, ACO took only up to a second longer than the  $\theta$ -heuristic, suggesting that it can feasibly be used on much larger graphs to reduce runtime as it can on smaller graphs.

## 5 Discussion

### 5.1 Interpretation of Results

On the tested graphs, ACO outperforms the  $\theta$ -heuristic on 17 out of 31 trials, and underperforms only 4 times. While ACO is significantly slower than the  $\theta$ -heuristic, sometimes taking over one hundred times as long, it often finds solutions 3-4x larger – for example, **DNC corecipient**, **moreno**, **nowikinews**, **matter**, and **wikitalnknds**. Additionally, in cases where ACO does outperform the  $\theta$ -heuristic, it results in significantly speed boosts to the brute-force algorithm.

Small improvements in ACO solutions can meaningfully reduce the time that Branch and Pivot takes. For example, FINISH.

### 5.2 Limitations

Our evaluation of ACO-seeded branch-and-pivot was constrained to 16 small graphs. Branch-and-pivot’s runtime grows exponentially with graph size even under a tight bound, and running it to completion on our larger test graphs

(e.g., `matter`, `dimacs-polblogs`) was not feasible on the hardware available for this study. As a result, our seeding experiment cannot yet confirm whether the pruning benefit observed on small graphs extends to the graphs where ACO’s advantage over the  $\theta$ -heuristic is largest (see Section 4).

However, it is likely that ACO seeding will scale to larger graphs. ACO’s measured runtime scales polynomially with graph size, and thus does not .

Additionally, we injected bicliques with  $\theta$  vertices on both sides into the graphs. [1] states that their proposed  $\theta$ -heuristic was inspired by  $k$ -MDB with exactly  $\theta$  vertices on one side and many more vertices on the other side. However, out of the 17 graphs where ACO beat the  $\theta$ -heuristic, on 8, the ACO solution had exactly  $\theta$  nodes on one side. This suggests that, even on graphs where the solution meets the condition for which the  $\theta$ -heuristic was designed, ACO can outperform its solution quality in certain instances.

## 6 Conclusion

Ant Colony Optimization is a promising heuristic approach to identifying large  $k$ -DB. As an emergent field, there do not exist many tested heuristics for this problem. The  $\theta$ -heuristic, the only previously proposed heuristic we could identify, is significantly faster than our algorithm but often does not produce solutions as large. While ACO takes longer, it can also function as a much more effective upper bound for brute-force search algorithms, cutting search time by as much as 98%.

### 6.1 Future Work

The graphs excluded from our seeded branch-and-pivot experiment are precisely the ones where we would expect the seeding benefit to be largest. On several large graphs — such as `dblp-cite`, `ciaodvd`, and `matter` — ACO found bicliques roughly 3–4 $\times$  larger by edge count than the  $\theta$ -heuristic, at a runtime cost of only a fraction of a second to roughly one second more (Table 1). Because branch-and-pivot’s pruning power depends on the tightness of its initial bound  $D^*$ , a substantially larger seed on these graphs should, in principle, prune far more of the search tree than the  $\theta$ -heuristic’s seed does, potentially yielding much larger runtime reductions than we observed on the small graphs in Section 4. We were unable to test this directly because branch-and-pivot’s runtime on graphs of this size exceeded the resources available for this study. A natural next step is to rerun the seeded comparison on higher-performance hardware to determine whether this relationship holds at scale.

Additionally, ACO could be adapted to have  $N$  subspecies that find  $N$  solutions in parallel by using a repulsion mechanism similar to that proposed for diversity in other ant colony variants [9], extending previous multi-colony parallelization strategies [10].

Code is available on <https://github.com/pooch63/aco-kmdb>.

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## 8 Appendix

### A ACO vs $\theta$ -heuristic

Below are the results of running ACO ants vs the  $\theta$ -heuristic on 10 different graphs.  $|U_G|$ ,  $|V_G|$ , and  $|E_G|$  represent the total number of  $U$  vertices,  $V$  vertices, and edges in the original graph, respectively.  $|U_R|$ ,  $|V_R|$ , and  $|E_R|$  represent the

total number of  $U$  vertices,  $V$  vertices, and edges in the reduced graph that both heuristics are run on, respectively.

ACO trials were run with 100 ants for a total of 5 iterations. ITB represents the number of iterations that were necessary for the ants to reach their largest solution. TTB represents the amount of time the ants required to finish their ITB – that is, to first find their best solution.

For both ACO and the  $\theta$ -heuristic,  $|U_{D^*}|$ ,  $|V_{D^*}|$ , and  $|E(D^*)|$  represent the number of  $U$  vertices,  $V$  vertices, and edges in their identified solutions. Times for both are shown in milliseconds.

ANALYSIS STILL LEFT TO DO: Vary  $k$  and  $\theta$ , even just doing  $k = 3$   $\theta = 6$ , to see if this holds up. Try pivot but with all these new graphs

OLD ACO, that doesn't use the candidate filtration method:

Table 3:  $k = 2, \theta = 5$ 

| Dataset           | $ U_G $ | $ V_G $ | $ E_G $ | $ U_R $ | $ V_R $ | $ E_R $ | ACO         |             |            |           |     |        | $\theta$ -Heuristic |             |            |      |
|-------------------|---------|---------|---------|---------|---------|---------|-------------|-------------|------------|-----------|-----|--------|---------------------|-------------|------------|------|
|                   |         |         |         |         |         |         | $ U_{D^*} $ | $ V_{D^*} $ | $ E(D^*) $ | Discovery | ITB | TTB    | $ U_{D^*} $         | $ V_{D^*} $ | $ E(D^*) $ | Time |
| Urwiki            | 221     | 1939    | 4018    | 14      | 45      | 168     | 5           | 5           | 23         | 1.7       | 1   | 1.7    | 5                   | 3           | 14         | 0.0  |
| Gnutella          | 4935    | 10856   | 40017   | 421     | 227     | 2758    | 6           | 7           | 40         | 242.0     | 3   | 61.3   | 5                   | 6           | 30         | 0.3  |
| Bitcoin           | 3286    | 3754    | 24209   | 1003    | 1105    | 15395   | 13          | 5           | 63         | 1266.5    | 5   | 255.7  | 5                   | 8           | 38         | 1.3  |
| Ciaodvd           | 1438    | 4299    | 40156   | 978     | 1593    | 35082   | 26          | 5           | 128        | 1613.8    | 4   | 238.9  | 5                   | 13          | 63         | 2.3  |
| Petster Hamster   | 2244    | 1766    | 16654   | 1310    | 852     | 13608   | 31          | 5           | 153        | 8803.5    | 3   | 563.2  | 5                   | 19          | 93         | 1.2  |
| Edit Guwikiquote  | 963     | 1678    | 11387   | 129     | 649     | 3328    | 5           | 37          | 183        | 730.5     | 1   | 25.9   | 5                   | 18          | 88         | 0.5  |
| Moreno Blogs      | 1065    | 990     | 19048   | 809     | 625     | 18044   | 9           | 32          | 286        | 962.8     | 4   | 129.9  | 5                   | 3           | 13         | 1.0  |
| Edit Vewiki       | 543     | 1532    | 11069   | 104     | 274     | 3385    | 6           | 80          | 478        | 325.7     | 2   | 23.3   | 5                   | 92          | 458        | 0.4  |
| Edit Zuwiktionary | 214     | 1404    | 7555    | 36      | 559     | 3142    | 5           | 118         | 588        | 467.7     | 1   | 24.9   | 5                   | 49          | 243        | 0.5  |
| Dnc Corecipient   | 653     | 714     | 12108   | 393     | 369     | 11379   | 35          | 41          | 1433       | 52.0      | 4   | 52.0   | 5                   | 89          | 443        | 1.0  |
| Nawiki            | 653     | 714     | 12108   | 393     | 369     | 11379   | 35          | 41          | 1433       | 53.2      | 4   | 53.2   | 5                   | 89          | 443        | 1.5  |
| P2P Gnutella09    | 3055    | 8038    | 26036   | 455     | 106     | 4056    | 4           | 4           | 14         | 14.1      | 2   | 14.1   | 5                   | 5           | 23         | 0.3  |
| Dblp Cite         | 3158    | 11566   | 49782   | 2466    | 2955    | 31279   | 5           | 6           | 28         | 5259.9    | 3   | 5259.9 | 5                   | 18          | 88         | 2.6  |
| Moreno Health     | 2313    | 2441    | 12992   | 325     | 275     | 1496    | 11          | 3           | 31         | 405.5     | 2   | 50.3   | 5                   | 6           | 29         | 0.2  |
| Pgp Arenas        | 5596    | 9657    | 24339   | 936     | 1247    | 9791    | 4           | 31          | 122        | 211.3     | 1   | 211.3  | 5                   | 27          | 133        | 1.0  |
| Dimacs Polblogs   | 1224    | 1224    | 33453   | 945     | 946     | 32524   | 10          | 23          | 228        | 586.7     | 4   | 212.7  | 5                   | 76          | 378        | 2.8  |
| Openflights       | 2918    | 2921    | 30523   | 1225    | 1222    | 25462   | 8           | 37          | 294        | 280.5     | 4   | 280.5  | 5                   | 67          | 333        | 2.7  |
| Matter            | 16726   | 22015   | 58618   | 2283    | 3276    | 11825   | 2           | 5           | 8          | 24900.0   | 3   | 7906.6 | 5                   | 4           | 18         | 1.3  |
| Music             | 100952  | 70519   | 130457  | 63      | 81      | 298     | 4           | 4           | 14         | 32.5      | 2   | 3.7    | 5                   | 3           | 13         | 0.1  |
| Wiki Talk Nds     | 605     | 23094   | 27455   | 45      | 149     | 616     | 4           | 4           | 14         | 238.0     | 1   | 5.3    | 5                   | 1           | 3          | 0.1  |
| Maayan Vidal      | 1633    | 2929    | 6749    | 81      | 134     | 547     | 3           | 8           | 22         | 2.4       | 1   | 2.4    | 5                   | 3           | 13         | 0.1  |
| Boxes             | 15237   | 1128    | 16239   | 14      | 14      | 50      | 5           | 5           | 23         | 8.0       | 1   | 0.4    | 5                   | 5           | 23         | 0.0  |
| Eswiki            | 48972   | 4896756 | 4896779 | 5       | 5       | 23      | 5           | 5           | 23         | 13.6      | 1   | 0.3    | 5                   | 5           | 23         | 0.0  |
| Euroroad          | 891     | 1092    | 1440    | 5       | 5       | 23      | 5           | 5           | 23         | 7.2       | 1   | 0.3    | 5                   | 5           | 23         | 0.0  |
| Facebook          | 28      | 2885    | 3003    | 9       | 8       | 36      | 5           | 5           | 23         | 4.4       | 1   | 0.4    | 5                   | 5           | 23         | 0.0  |
| Jawiki            | 31009   | 3100497 | 3100520 | 5       | 5       | 23      | 5           | 5           | 23         | 6.8       | 1   | 0.3    | 5                   | 5           | 23         | 0.0  |
| Moreno            | 888     | 1597    | 2300    | 13      | 12      | 50      | 5           | 5           | 23         | 7.6       | 1   | 0.4    | 5                   | 5           | 23         | 0.0  |
| Appliances        | 1755732 | 104237  | 2128628 | 1289    | 421     | 4175    | 17          | 3           | 49         | 1811.6    | 2   | 425.1  | 5                   | 3           | 13         | 0.4  |
| Edit Srwikiquote  | 402     | 1823    | 8695    | 80      | 396     | 2637    | 5           | 25          | 123        | 448.2     | 1   | 11.2   | 5                   | 25          | 123        | 0.4  |
| Kuwiki            | 219     | 1094    | 4951    | 45      | 248     | 1668    | 5           | 58          | 288        | 219.6     | 1   | 6.8    | 5                   | 58          | 288        | 0.3  |
| Edit Mrwiktionary | 281     | 3995    | 14375   | 46      | 1014    | 4630    | 5           | 94          | 468        | 1328.5    | 1   | 54.7   | 5                   | 94          | 468        | 0.7  |

NEW ACO, that DOES use the candidate filtration method:

Table 4:  $k = 2, \theta = 5$ 

| Dataset           | $ U_G $ | $ V_G $ | $ E_G $ | $ U_R $ | $ V_R $ | $ E_R $ | ACO         |             |            |           |     |       | $\theta$ -Heuristic |             |            |      |
|-------------------|---------|---------|---------|---------|---------|---------|-------------|-------------|------------|-----------|-----|-------|---------------------|-------------|------------|------|
|                   |         |         |         |         |         |         | $ U_{D^*} $ | $ V_{D^*} $ | $ E(D^*) $ | Discovery | ITB | TTB   | $ U_{D^*} $         | $ V_{D^*} $ | $ E(D^*) $ | Time |
| Appliances        | 1755732 | 104237  | 2128628 | 1289    | 421     | 4175    | 5           | 5           | 23         | 84.3      | 1   | 13.3  | 5                   | 3           | 13         | 0.7  |
| Maayan Vidal      | 1633    | 2929    | 6749    | 81      | 134     | 547     | 5           | 5           | 23         | 49.0      | 1   | 2.2   | 5                   | 3           | 13         | 0.2  |
| Music             | 100952  | 70519   | 130457  | 63      | 81      | 298     | 5           | 5           | 23         | 9.9       | 1   | 1.5   | 5                   | 3           | 13         | 0.1  |
| Urwiki            | 221     | 1939    | 4018    | 14      | 45      | 168     | 5           | 5           | 23         | 28.3      | 1   | 1.9   | 5                   | 3           | 14         | 0.1  |
| Wiki Talk Nds     | 605     | 23094   | 27455   | 45      | 149     | 616     | 5           | 7           | 33         | 10.7      | 1   | 10.7  | 5                   | 1           | 3          | 0.2  |
| Moreno Health     | 2313    | 2441    | 12992   | 325     | 275     | 1496    | 6           | 6           | 34         | 107.5     | 2   | 9.2   | 5                   | 6           | 29         | 0.4  |
| Gnutella          | 4935    | 10856   | 40017   | 421     | 227     | 2758    | 10          | 7           | 68         | 15.4      | 3   | 15.4  | 5                   | 6           | 30         | 0.4  |
| Bitcoin           | 3286    | 3754    | 24209   | 1003    | 1105    | 15395   | 11          | 7           | 75         | 102.2     | 5   | 102.2 | 5                   | 8           | 38         | 2.1  |
| Matter            | 16726   | 22015   | 58618   | 2283    | 3276    | 11825   | 9           | 9           | 79         | 363.0     | 2   | 88.5  | 5                   | 4           | 18         | 2.3  |
| Ciaodvd           | 1438    | 4299    | 40156   | 978     | 1593    | 35082   | 18          | 8           | 142        | 663.3     | 3   | 72.7  | 5                   | 13          | 63         | 3.5  |
| Edit Nowikinews   | 430     | 4060    | 16923   | 82      | 1192    | 5177    | 5           | 33          | 163        | 79.8      | 1   | 79.8  | 5                   | 9           | 43         | 1.3  |
| Edit Guwikiquote  | 963     | 1678    | 11387   | 129     | 649     | 3328    | 5           | 37          | 183        | 42.3      | 1   | 42.3  | 5                   | 18          | 88         | 0.8  |
| Pgp Arenas        | 5596    | 9657    | 24339   | 936     | 1247    | 9791    | 13          | 17          | 219        | 204.2     | 5   | 96.8  | 5                   | 27          | 133        | 1.6  |
| Moreno Blogs      | 1065    | 990     | 19048   | 809     | 625     | 18044   | 24          | 11          | 262        | 474.0     | 4   | 76.4  | 5                   | 3           | 13         | 1.7  |
| Petster Hamster   | 2244    | 1766    | 16654   | 1310    | 852     | 13608   | 70          | 5           | 348        | 17.1      | 1   | 17.1  | 5                   | 19          | 93         | 2.0  |
| Edit Zuwiktionary | 214     | 1404    | 7555    | 36      | 559     | 3142    | 5           | 118         | 588        | 769.8     | 1   | 42.9  | 5                   | 49          | 243        | 0.8  |
| Dnc Corecipient   | 653     | 714     | 12108   | 393     | 369     | 11379   | 38          | 38          | 1442       | 66.2      | 4   | 66.2  | 5                   | 89          | 443        | 1.7  |
| Nawiki            | 653     | 714     | 12108   | 393     | 369     | 11379   | 38          | 38          | 1442       | 59.8      | 4   | 59.8  | 5                   | 89          | 443        | 1.6  |
| Edit Jamwiki      | 177     | 2741    | 17669   | 64      | 1484    | 8298    | 6           | 268         | 1606       | 3549.9    | 1   | 342.9 | 5                   | 184         | 918        | 1.8  |
| P2P Gnutella09    | 3055    | 8038    | 26036   | 455     | 106     | 4056    | 7           | 4           | 26         | 84.3      | 1   | 5.4   | 5                   | 5           | 23         | 0.5  |
| Edit Rowikibooks  | 443     | 4221    | 18549   | 69      | 425     | 1515    | 4           | 9           | 34         | 59.7      | 1   | 14.8  | 5                   | 5           | 23         | 0.3  |
| Dimacs Polblogs   | 1224    | 1224    | 33453   | 945     | 946     | 32524   | 16          | 17          | 270        | 356.2     | 5   | 116.9 | 5                   | 76          | 378        | 4.3  |
| Openflights       | 2918    | 2921    | 30523   | 1225    | 1222    | 25462   | 12          | 24          | 286        | 492.4     | 3   | 76.6  | 5                   | 67          | 333        | 4.3  |
| Edit Vewiki       | 543     | 1532    | 11069   | 104     | 274     | 3385    | 9           | 40          | 358        | 68.6      | 5   | 68.6  | 5                   | 92          | 458        | 0.7  |
| Boxes             | 15237   | 1128    | 16239   | 14      | 14      | 50      | 5           | 5           | 23         | 4.0       | 1   | 0.6   | 5                   | 5           | 23         | 0.0  |
| Eswiki            | 48972   | 4896756 | 4896779 | 5       | 5       | 23      | 5           | 5           | 23         | 3.5       | 1   | 0.6   | 5                   | 5           | 23         | 0.0  |
| Euroroad          | 891     | 1092    | 1440    | 5       | 5       | 23      | 5           | 5           | 23         | 6.2       | 1   | 0.6   | 5                   | 5           | 23         | 0.0  |
| Facebook          | 28      | 2885    | 3003    | 9       | 8       | 36      | 5           | 5           | 23         | 6.8       | 1   | 0.6   | 5                   | 5           | 23         | 0.0  |
| Jawiki            | 31009   | 3100497 | 3100520 | 5       | 5       | 23      | 5           | 5           | 23         | 7.7       | 1   | 0.6   | 5                   | 5           | 23         | 0.0  |
| Moreno            | 888     | 1597    | 2300    | 13      | 12      | 50      | 5           | 5           | 23         | 7.4       | 1   | 0.7   | 5                   | 5           | 23         | 0.0  |
| Edit Mkwikisource | 218     | 6635    | 13824   | 19      | 121     | 399     | 4           | 9           | 34         | 32.1      | 1   | 5.2   | 5                   | 2           | 8          | 0.1  |
| Dblp Cite         | 3158    | 11566   | 49782   | 2466    | 2955    | 31279   | 5           | 18          | 88         | 186.1     | 3   | 186.1 | 5                   | 18          | 88         | 4.7  |
| Edit Srwikiquote  | 402     | 1823    | 8695    | 80      | 396     | 2637    | 5           | 25          | 123        | 12.2      | 1   | 12.2  | 5                   | 25          | 123        | 0.6  |
| Kuwiki            | 219     | 1094    | 4951    | 45      | 248     | 1668    | 5           | 58          | 288        | 297.3     | 1   | 15.3  | 5                   | 58          | 288        | 0.4  |
| Edit Mwiktionary  | 281     | 2005    | 14275   | 46      | 1014    | 4620    | 5           | 94          | 468        | 1261.7    | 1   | 94.6  | 5                   | 94          | 468        | 1.1  |



## B Branch and Pivot Pseudocode

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**Algorithm 4** Branch and Pivot
 

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 $D^* \leftarrow \text{THETAHEURISTIC}(G)$ 
function BRANCHP( $S = (U_S, V_S), C = (U_C, V_C)$ )
  if  $C = \emptyset$  then
    if  $|E(S)| > |E(D^*)|$  and  $|U_S| \geq \theta$  and  $|V_S| \geq \theta$  then
       $D^* \leftarrow S$ 
    return
   $C_0 \leftarrow \{v \in C \mid \bar{d}_S(v) = 0\}$ 
  if  $C_0 = \emptyset$  or  $|C \setminus C_0| \geq k - |\bar{E}(S)|$  then
     $u \leftarrow \operatorname{argmax}_{v \in C} \bar{d}_S$ 
     $(S', C') \leftarrow \text{UPDATE}(S, C, u)$ 
    BRANCHP( $S' \cup \{u\}, C'$ )
    BRANCHP( $S, C \setminus \{u\}$ )
  else
     $u \leftarrow \operatorname{argmin}_{v \in C_0} \bar{d}_C$ 
    if  $\bar{d}_C(u) > k - |\bar{E}(S)| > 0$  then
       $(S', C') \leftarrow \text{UPDATE}(S, C, u)$ 
      BRANCHP( $S' \cup \{u\}, C'$ )
      BRANCHP( $S, C \setminus \{u\}$ )
    else
      for  $v \in \{u\} \cup N_C(u)$  do
         $(S', C') \leftarrow \text{UPDATE}(S, C, v)$ 
        BRANCHP( $S' \cup \{v\}, C'$ )
         $C \leftarrow C \setminus \{v\}$ 

```

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$D^*$  is the  $k$ -MDB of the graph.

We varied ACO configurations across two settings: whether an ant prefers the smaller side (P), and whether it restricts its search to just the neighbors of its most recently visited node (N). We find that ACO-PN outperforms all other configurations on runtime, and does not meaningfully sacrifice solution quality.

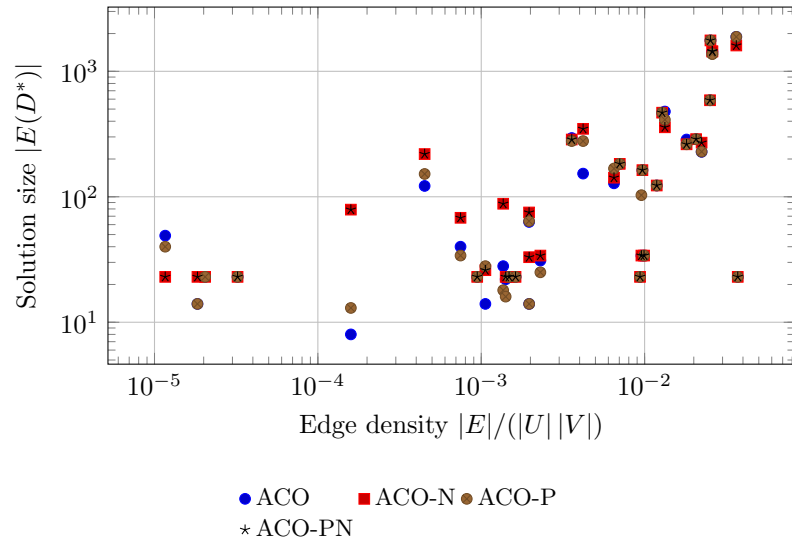


Figure 3: ACO solution size vs. graph edge density (flag settings).

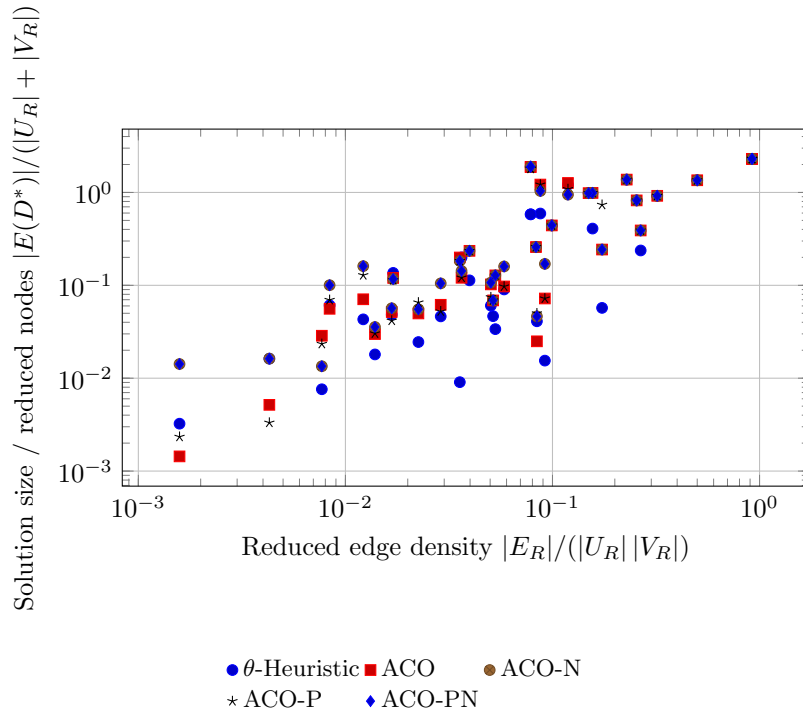


Figure 4: Solution size per reduced node vs. reduced edge density ( $\theta$ -heuristic and ACO flag settings).

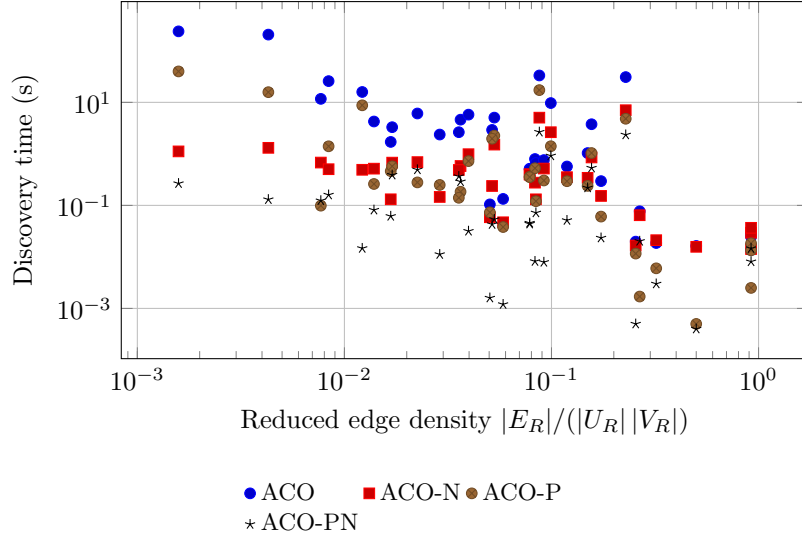


Figure 5: ACO discovery time vs. reduced edge density (flag settings).

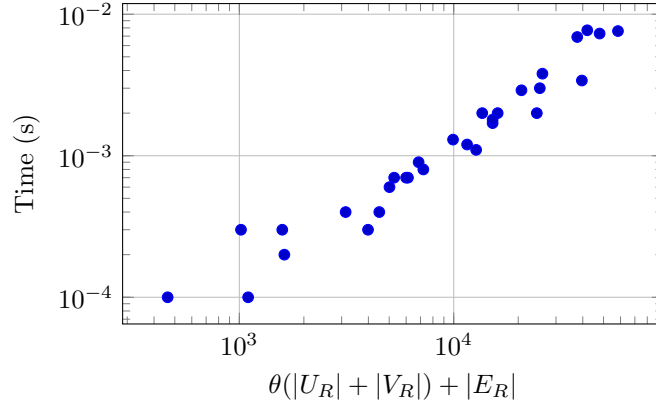


Figure 6:  $\theta$ -heuristic wall time vs.  $\theta(|U_R| + |V_R|) + |E_R|$ .